# LAB 2: SUBGRADIENT METHOD AND PROXIMAL GRADIENT DESCENT

**Step 1.** Generate the data as in:
https://www.cvxpy.org/examples/machine_learning/lasso_regression.html

**Step 2.** Derive the subgradient algorithm to solve $l_1$-regularized least-square problem.

$$\min.\left(\frac{1}{m}\right)\left|\left|X\beta - Y\right|\right|_2^2 + \lambda\left|\left|\beta\right|\right|_1.$$

where $\lambda > 0$.

**Step 3.** Write the Python program to solve the $l_1$-regularized least-square problem with **subgradient method** with fixed stepsize and diminised stepsize. Do they converge to the optimal solution yielded by step 1?

**Step 4.** Derive the proximal algorithm to solve the above problem.

**Step 5.** Write the Python program to solve the $l_1$-regularized least-square problem with **proximal gradient descent.** Does it converge to the optimal solution yielded by step 1?

**Step 6.** Write the Python program to solve the $l_1$-regularized least-square problem with **stochastic proximal gradient descent** with different mini-batch sizes**.**

**Step 7.** Compare results from above methods.

**References:**

- https://www.cvxpy.org/examples/machine_learning/lasso_regression.html

**Rubrics for lab assessment:**

| **Criteria** | 8-10 | 5-7 | 4-0 |
|---|---|---|---|
| **Codes (60 points)** | The codes run correctly for all steps, the iterative algorithms converge to the optimal solution;<br>Plots demonstrate the results clearly | 1-2 steps haven't been finished;<br>The code is correct, however, the algorithms do not converge and some plots are in correct. | Just few steps have been done. |
| **Presentation (40 points)** | Demonstrates full knowledge of the topic, the report is well organized | The content shows some understandin g and comprehension, few steps haven't been finished | The presentation has some information about the topic, but is mostly based on clichés and basic knowledge. |